

Stable VT with ICD Firing



Section I: Scenario Demographics

Scenario Title:	Stable VT with ICD Firing		
Date of Development:	23/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	Expose learners to a rare presentation that may trigger discomfort among healthcare workers.
CRM Objectives:	Communicate effectively with team members and reassure them regarding their safety when ICD fires.
Medical Objectives:	1) Appropriately call for magnet application in order to terminate ICD firing. 2) Perform appropriate initial workup for misfiring ICD. 3) Administer appropriate antiarrhythmic therapy for stable VT. 4) Administer second line antiarrhythmic therapy for stable VT and seek expert consultation.

Case Summary: Brief Summary of Case Progression and Major Events

A 40-year-old male presents to the ED complaining that his ICD keeps firing. He will have a HR of 180 and VT on the monitor. He will occasionally yell "ow." The team will need to work through medical management of VT, while considering magnet placement for patient comfort. The patient will remain stable but will trigger VT with his agitation.

Facilitators Required to Run Session

Instructors (faculty or senior resident): 1-2 (one to observe, one to run mannequin if no sim tech)
Confederate nurse: 1 (to assist at bedside, question team re: safety of touching patient)
Sim tech: 1

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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
Nurse	Question team leader as to safety of touching patient with ICD firing. Cue team to patient's status PRN (including agitation)		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required. (Can place something under mannequin skin on chest to resemble implanted ICD.)			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 40-year-old male presents to your tertiary care ED complaining that his ICD keeps firing. He keeps yelling "ow" and jumping/jerking every couple minutes during his triage. He has an ICD in place because he had previous myocarditis that left him with a poor EF.

B. Patient Profile and History

Patient Name: Gerald Jones	Age: 40	Weight: 70 kg
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Gender: ☒ M ☐ F	Code Status: Full
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Chief Complaint: ICD keeps firing

History of Presenting Illness: For the last 20 minutes, ICD keeps firing. Patient is getting very frustrated with how uncomfortable the shocks are.

Past Medical History:	Recurrent VT ICD placed 3 years ago for medication-refractory VT	Medications:	Amiodarone 400mg daily
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Allergies: None.

Social History: Non-smoker. Denies recreational drugs.

Family History: N/A.

Review of Systems:	CNS:	Nil.
	HEENT:	Nil.
	CVS:	Hurts when ICD fires. No preceding CP. No palpitations.
	RESP:	No preceding SOB or SOBOE.
	GI:	Nil.
	GU:	Nil.
	MSK:	Nil.
	INT:	

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
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HR: 180/min	BP: 100/60	RR: 16/min	O ₂ SAT: 96% RA
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Rhythm: VT	T: 36.5°C	Glucose: 7.1 mmol/L	GCS: 15 (E 4 V5 M6)
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General Status: Alert. **Appears very uncomfortable when ICD goes off. Screams very loudly.**

CNS:	Alert, oriented. Agitated and begging team to make ICD stop firing.
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HEENT:	Nil.
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CVS:	No murmur. Pulse palpable at 180. No signs CHF on exam.
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RESP:	GAEB. No adventitious. No signs CHF.
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ABDO:	Soft, NT.
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GU:	Nil.
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MSK:	Normal.	SKIN:	Palpable ICD to left upper chest.
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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: HR: 180/min BP: 100/60 RR: 16/min O ₂ SAT: 96% RA T: 36.5°C	Alert and oriented. Stable but very agitated about ICD shocks. Begging MD to make it stop.	<u>Learner Actions</u> ☐ Apply monitors, full vitals ☐ Obtain IV access ☐ Apply defibrillator pads ☐ Full hx/px ☐ ECG ☐ Labs (including extended electrolytes, troponin) ☐ Apply magnet ☐ Amiodarone 150mg iv (or lidocaine 1.5mg/kg iv)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Magnet applied → Patient stops screaming - RN asks if safe to touch patient → If not reassured, RN will not touch patient - If plans for sedation/external defib, patient to say "can't you just unhook this thing instead?" <u>Triggers</u> <i>For progression to next state</i> - 5 minutes → 2. Persistent VT
2. Persistent VT Unchanged.	Patient more calm (if magnet applied). If not, patient unchanged.	<u>Learner Actions</u> ☐ Second dose amiodarone ☐ Call arrhythmia/cardiology (will suggest giving procainamide if not yet given) ☐ Procainamide 20-50mg/min until rhythm stops ☐ Magnesium sulfate 2g iv	<u>Triggers</u> - 2 min after procainamide given → 3. VT resolution - 8 minutes → 4. Patient agitated
3. VT resolution Rhythm → Sinus HR → 95 BP → 110/70		<u>Learner Actions</u> ☐ Call arrhythmia ☐ Consider procainamide infusion ☐ CXR re: lead placement	<u>Triggers</u> - 1 min into state → 4. Patient agitated
4. Patient agitated Rhythm: VT HR: 180/min BP: 150/90	Patient gets very upset about needing to be in hospital and triggers VT.	<u>Learner Actions</u> ☐ Consider benzodiazepine administration ☐ Provide reassurance ☐ Restart procainamide (if stopped) ☐ Magnesium sulfate 2g iv (if not yet given) ☐ 3 rd dose amiodarone ☐ Consider esmolol 500mcg/kg bolus then 50mcg/kg/min infusion	<u>Modifiers</u> - Reassurance of patient by dedicated team member → Patient calms down <u>Triggers</u> - Benzo given → 5. Resolution - 12 minutes → 5. Resolution
5. Resolution Rhythm → Sinus HR → 95 BP → 110/70	Patient calm.	<u>Learner Actions</u> ☐ Call arrhythmia ☐ Amiodarone infusion or procainamide infusion	END CASE 1 minute into state → Arrhythmia arrives

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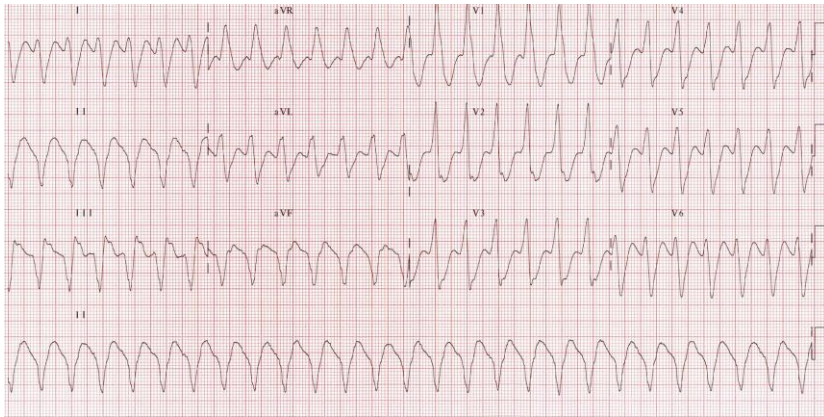
Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

None required for case.

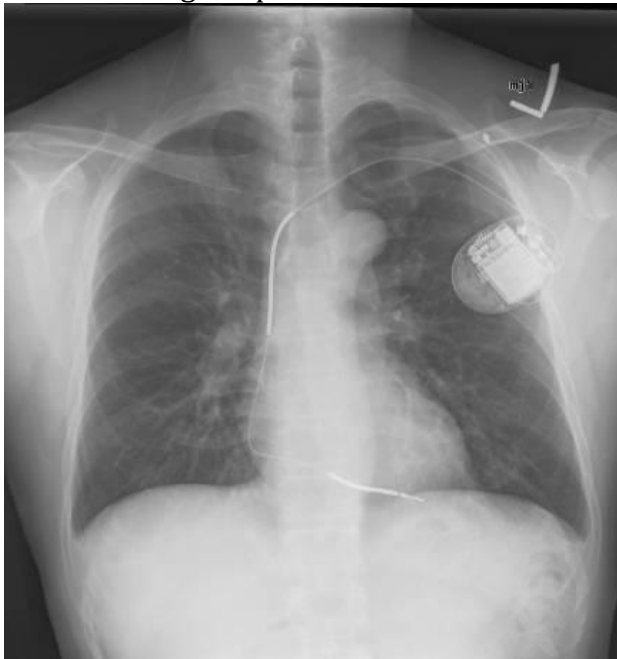
Images (ECGs, CXRs, etc.)

ECG – VT



ECG source: <http://lifeinthefastlane.com/ecg-library/ventricular-tachycardia/>

CXR – ICD in good position



CXR source: https://commons.wikimedia.org/wiki/File:Implantable_cardioverter_defibrillator_chest_X-ray.jpg

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How did it feel to have your colleague questioning your judgment around the safety of touching the patient? Do you have any strategies for dealing with conflict among coworkers? 2) What is your approach to the management of stable VT? What antidysrhythmic options are there? 3) What effect does placing a magnet have for an ICD? For a pacemaker? 4) Does ACLS management change in a patient with an ICD?	
Key Moments	
Acknowledging safety of touching patient.	
Working through antidysrhythmic treatment options.	
Management of catecholamine surge.	